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Review

Urban Air Mobility Communications and Networking: Recent Advances, Techniques, and Challenges

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Abstract: Over the past few years, our traditional ground-based transportation system has encountered various challenges, including overuse, traffic congestion, growing urban populations, high infrastructure costs, and disorganization. Unmanned aerial vehicles, commonly referred to as drones, have significantly impacted aerial communication in both the academic and industrial sectors. Therefore, researchers and scientists from the aviation and automotive industries have collaborated to create an innovative air transport system that solves traditional transport problems. In the coming years, urban air mobility (UAM) is expected to become an emerging air transportation system that enables on-demand air travel. UAM is also anticipated to offer more environmentally friendly, cost-effective, and faster modes of transportation than ground-based alternatives. Owing to the unique characteristics of personal air vehicles, ensuring reliable communication and maintaining proper safety and security, air traffic management, collision detection, path planning, and highly accurate localization and navigation have become increasingly complex. This article provides an extensive literature review of recent technologies to address the challenges UAM faces. First, we present UAM communication requirements in terms of coverage, data rate, latency, spectrum efficiency, networking, and computing capabilities. Subsequently, we identify the potential key technological enablers to meet these requirements and overcome their challenges. Finally, we discuss open research issues, challenges, and future research directions for UAM deployment.

Keywords: advanced air mobility (AAM); drones; eVTOL; flying vehicles; future mobility; personal air vehicles (PAVs); urban air mobility (UAM); unmanned aerial vehicles (UAVs); vertical takeoff and landing

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1. Introduction

As urban population density has increased in recent decades, traffic congestion and commuter delays have risen unprecedentedly. By 2050, approximately 80% of the European and North American populations are estimated to live in metropolitan regions [1]. As a result of this phenomenal growth, major cities worldwide continue to face escalating infrastructure and mobility issues. Additionally, urbanization leads to conflict, pollution, and congestion, negatively affecting both human and environmental health [2]. According to the European Institute of Innovation and Technology, congestion costs 130 billion euros annually in Europe [3]. Therefore, it is necessary to investigate new mobility concepts and paradigms to reduce commuting times, avoid ground traffic, and enable point-to-point flight between cities. A revolutionary air transportation method called urban air mobility (UAM) has emerged to alleviate ground traffic saturation by providing an alternative mode of transport for both intercity and intracity travel. In this scenario, UAM offers a new dimension to city commuting by extending transportation into the urban airspace. With UAM, intercity connections are enhanced through vertical mobility, thus offering efficient and environmentally friendly alternatives to traditional ground transport. Transport authorities expect UAM to be a crucial part of the smart city initiative. In particular, municipalities anticipate that UAM will deliver a variety of services, including medical assistance, rescue missions, and parcel delivery, as part of a smart city initiative [4]. As shown in Figure 1, UAM has a wide range of applications in both the military and civil sectors.

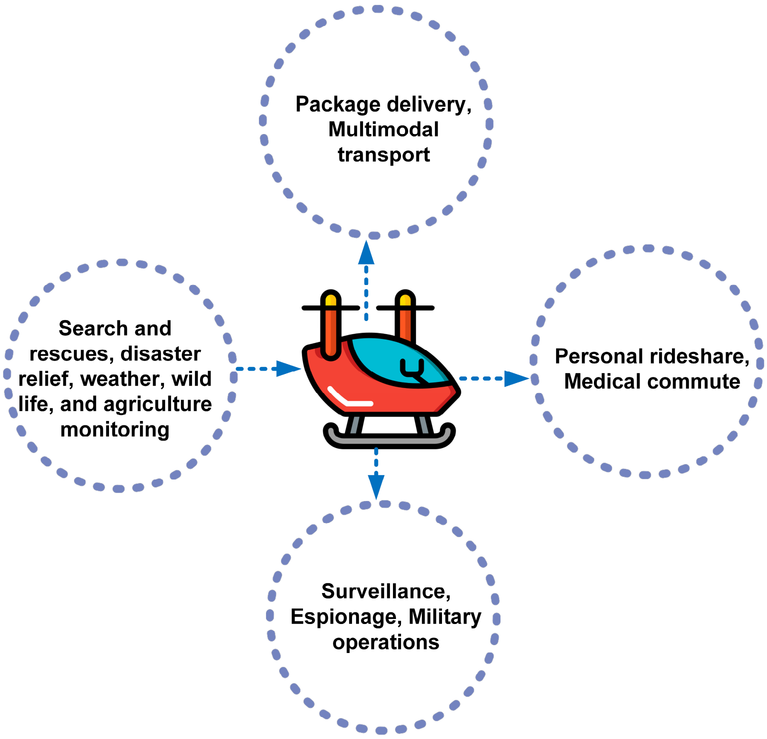

Figure 1. Applications of UAM in the military and civil domains.

With the advent of UAM, the aviation industry has revolutionized and disrupted traditional mobility systems and urban planning. Leading aviation companies are competing to advance the vertical takeoff and landing (VTOL) technologies and prototypes and the mass-production of VTOL vehicles that air taxi services will use in the future [5]. An air taxi is a small electric aircraft with an average seating capacity of four that can be used on demand to transport passengers in metropolitan areas. In addition, several companies have already conducted extensive test flights using their electrical vertical takeoff and landing (eVTOL) technology, which has been demonstrated in several countries worldwide [6]. In a large city, air taxis can take off and land vertically using eVTOL technology, which is similar to the technology used by helicopters. UAM primarily utilizes VTOL technology to develop aircraft, providing an eco-friendly solution in terms of pollution, economic viability, enhanced safety, and improved fault tolerance. Most VTOL aircraft are likely to be powered by eVTOL technology, enabling operations without a runway, thereby providing significant savings in both space and time [7]. UAM still needs to be available and usable on demand because many challenges remain to be overcome [8]. UAM operation and connectivity are challenging issues. Owing to its low-altitude flight characteristics, UAM experiences interference and disturbances in communication with base stations [9]. Security is of paramount importance in the communication and control paradigms. During takeoff and landing, UAM requires congestion management and air traffic routing, which are far more complex and prone to accidents than the traditional high-altitude flights [10]. Moreover, multi-eVTOL swarms present significant challenges in terms of collision avoidance, topology control, and task scheduling [11]. The effective and robust use of UAM requires a robust and scalable infrastructure, particularly in highly congested urban environments [12].

UAM represents a disruptive technology that has revolutionized the entire transportation industry. We provide a comparative analysis between UAM and existing transportation systems to better understand its feasibility. In recent years, unmanned aerial vehicles

(UAVs), commonly known as drones, have been operated with radio remote-control equipment and self-contained programmed controls, or with onboard computers that operate fully autonomously or intermittently [13]. A flying ad hoc network (FANET) consists of a variety of aerial nodes such as drones or flying vehicles, operating in the on-demand ad hoc mode [14]. A FANET can be utilized in a wide variety of applications, including search and rescue in remote areas, damage assessment, wildlife monitoring, the detection of forest fires, pollution or crowd monitoring, surveillance, and public safety [15]. UAV or drone networks are organized similarly to FANET [16]. However, the nodes consist solely of UAVs that are remotely controlled or have predetermined trajectories. The deployment of a drone- or UAV-based network requires precise orchestration of the UAVs' mobility and trajectory, which are essential characteristics of these networks [17]. The versatility, mobility, and flexibility of UAV or drone networks make them increasingly useful across various industries and fields. There is a wide range of applications, such as logistics delivery, agriculture, crop monitoring, border patrolling, planting and spraying, and temporary network deployment [18]. A UAM is a self-organizing architecture model divided into two main layers: aerial and ground. In urban environments, UAM refers to both manned and unmanned air vehicle systems used for the transportation of people and goods. With UAM, a new way of traveling within urban areas will be introduced, which will, in turn, help reduce congestion [19]. Table 1 summarizes the major characteristics of FANETs, UAV/drone networks, and UAM.

Table 1. Comparison of the main characteristics of FANETs, UAV/drone networks, and UAM.

Criteria	FANETs	UAV/Drone Networks	UAM
Domain	Temporary ad hoc network	Single- or multiuser operations	Specific missions
Autonomy level	Unmanned and fully autonomous	Unmanned	Manned (human-controlled) and unmanned (fully autonomous)
Control	Distributed	Distributed and centralized	Mostly centralized
Communication and navigation technology	Wi-Fi, GPS, and radio frequency (RF)	Wi-Fi, GPS, and RF	Fourth-generation wireless (4G)/fifth-generation wireless (5G), GPS, RF, and visual signal
Path planning	Dynamic and based on the applications	Depends on the remote controller	Pre-planned path for specific mission
Communication distance	Short range	Short and medium range	Long range
Flying altitude	Low to medium	Low	Low to medium
Propulsion technology	Battery power	Battery power	Battery power and fuel
Network integration	Satellites and high-altitude platform stations (HAPs).	Aerial-to-ground (A2G) station	Air traffic control on a ground station

UAM technology includes advanced aircraft, upgraded airspace management systems, and autonomous navigation systems. UAM is also supported by a range of advanced technologies such as electric propulsion, high-tech materials, and sensors for navigation and safety [20]. These technologies are used to develop quieter, safer, and more efficient vehicles than traditional aircraft. The most prominent UAM projects include eVTOL aircraft, which take off vertically and fly horizontally [21]. These aircraft are designed to use electric propulsion and batteries, allowing them to be more efficient and quieter than traditional aircraft [22]. They will also be able to fly at lower altitudes, allowing them to access airspaces not currently used by commercial aircraft [23]. One of the key challenges for achieving success in UAM is maintaining safe flight in a highly dynamic environment. At present, most UAM research is focused on the structural aspect, and is similar to current air traffic management (ATM) techniques [24]. However, significant attention must be paid to UAM safety management. One major concern is the uncertainty of aircraft location. Due to positioning errors caused by the global positioning system (GPS), aircraft may not be able to accurately obtain the location information of other aircraft, leading to potential safety risks [25]. Furthermore, these issues can arise from inaccurate sensor data, performance variance caused by the dynamic nature of the environment, and environmental factors, such as winds, low-altitude birds, and high-rise buildings [26]. Therefore, more real-life constraints must be considered when designing a collision avoidance approach for UAM [27]. Although UAM is still in its early stages of development, its potential is immense. As technology continues to evolve, it could have a significant impact on urban transportation.

1.1. Contributions of This Study

A brief overview of UAM operations is provided in this article along with a discussion of their key features. These aspects include power consumption, vertical takeoff and landing systems, autonomy level, UAM integration with existing networks, and ATM. Subsequently, we review existing communication techniques in UAM. These techniques include communication data types, spectrum and frequency management, communication reliability, security issues, navigation, route networks, trajectory, collision avoidance, and flight scheduling. Finally, a discussion is presented on the opportunities and challenges associated with UAM development and its potential impact on society. Table 2 provides a list of acronyms for ease of reading and understanding.

Table 2. List of acronyms.

UAM—Urban air mobility	VTOL—Vertical takeoff and landing
eVTOL—Electrical vertical takeoff and landing	FANET—Flying ad hoc network
UAV—Unmanned aerial vehicle	ATM—Air traffic management
HAPs—High-altitude platform stations	A2G—Air-to-ground
A2A—Air-to-air	U2I—UAV-to-infrastructure
DEP—Distributed electric propulsion	UAS—Unmanned aerial aircraft
LOS—Line-of-sight	NLOS—Non-line-of-sight
BLOS—Blind-line-of-sight	CNPC—Control and non-payload communication
C2—Control and command	ATC—Air traffic control
UTM—Unmanned aircraft traffic management	GS—Ground station
LEO—Low-Earth orbit	GEO—Geostationary equatorial orbit
CDAA—Collision detect-and-avoid systems	CNS—Communication, navigation, and surveillance
D2D—Device-to-device	FAA—Federal aviation administration
EASA—European aviation safety agency	PAV—Personal air vehicle
SESAR—Single European Sky ATM research	NASA—National aeronautics and space administration
ACAS—Airborne collision avoidance system	TCAS—Traffic collision avoidance system

In this study, we aim to gain insight into UAM communication and networking techniques from the existing literature. The papers were retrieved from large online databases rather than being handpicked by specific publishers. As a part of our literature review, we searched for articles that address any aspect of UAM communication and networking. The most significant technical publishers, such as the association for computing machinery (ACM), Elsevier, IEEE Explore, Springer, Frontiers, Google Scholar, and arXiV, were searched for relevant publications. In this study, we gathered articles published in 2015 as well as the most recent papers published in 2024. To accommodate the varying capabilities of the search functions of each database, we made necessary adjustments to the search string. Since UAM is a relatively new technology, most of the references are current.

1.2. Outline of the Paper

This review is organized into five sections, as shown in Figure 2. The remainder of the paper is outlined below. In the next section, we present the different aspects of UAM operating concepts, including the power source, takeoff and landing, level of autonomy, integration with existing networks, and energy consumption. Afterward, we reviewed recent communication techniques in UAM. Following this, we present open research issues, challenges, and future research directions. Finally, we conclude the study in the next section.

Section 1: Introduction	
1.1 Contribution of this study 1.2. Outline of the paper	
Section 2: Operational concepts of UAM	
2.1. Power source 2.2. Takeoff and landing 2.3. Level of autonomy	2.4. Integration with existing networks 2.5. Air traffic management (ATM) 2.6. Energy efficiency
Section 3: Communication techniques in UAM	
3.1 Data type 3.2 Spectrum and carrier frequencies 3.3 Reliable communications 3.4 Data rate and latency 3.5 Communication zone 3.6 Secure communications 3.7 Network topology 3.8 Navigation and localization 3.9 Collision detect-and-avoid (CDAA) systems 3.10 Routing 3.11 Path planning and mission planning 3.12 Flight scheduling	
Section 4: Opportunities and challenges	
Section 5: Conclusion	

Figure 2. Outline of the paper.

2. Operational Concepts of UAM

Generally, UAM specializes in transportation within urban areas or intercity within short and medium distances (3–100 km), flying at ultralow altitudes below 100 m, and in low-altitude airspace between 100 m and 1000 m [28]. With the current technology, manned aerial vehicles are capable of carrying up to five passengers. In urban flights, batteries (lithium or hydrogen fuel cells) provide power, whereas in intercity flights, hybrid-electric propulsion may be employed [29]. Recent technological advancements in electrification, automation, and VTOL have made it possible to develop new aircraft designs, services, and business models. The combination of these factors provides new opportunities for the on-demand use of UAM aircraft in the transportation of commodities. UAM vehicles have many different requirements and boundary conditions compared with conventional aircraft designs [30]. In Figure 3, the fundamental design factors that influence the selection of aircraft type and the ensuing design specifications are described.

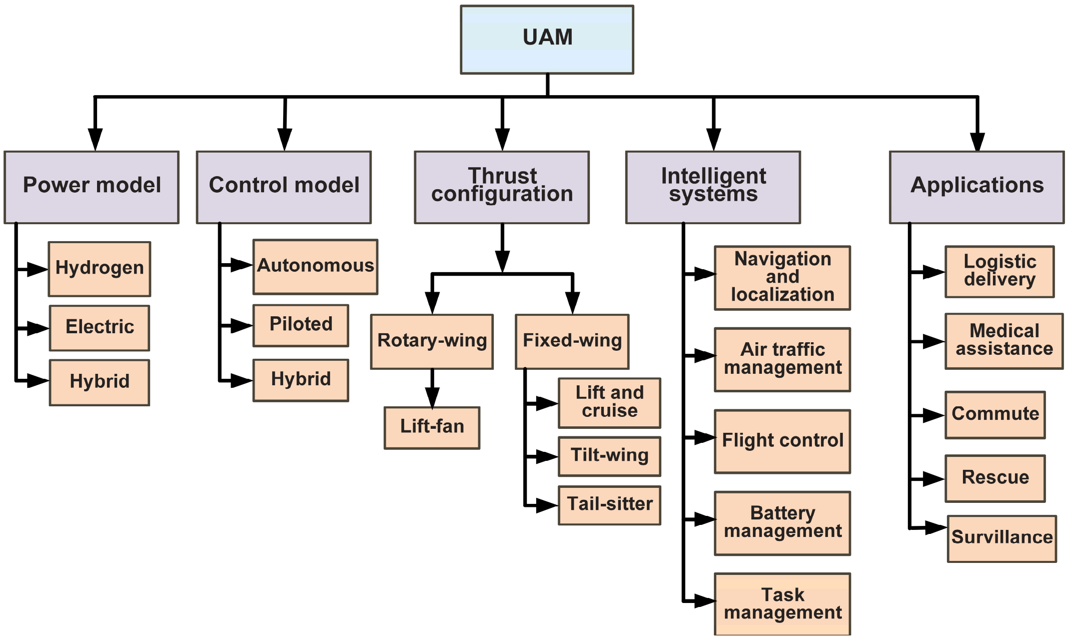

Figure 3. Taxonomy of various aspects of UAM.

2.1. Power Source

Recent developments in distributed electric propulsion (DEP) and advanced electric technology have led to an increase in the popularity of UAM aerial vehicles [31]. With the help of these technologies, we have been able to combine strongly divergent and underlying operational needs to design different aircraft types. As shown in Figure 3, UAMs can be divided into two categories based on the mechanism that allows VTOL and lift generation during a flight. Through the development and use of DEP, academics and industry specialists have been able to design aircraft with acceptable levels of system complexity and weight for both rotary-wing and fixed-wing cruise aircraft [32]. Fuel combustion engines, particularly diesel engines, offer a much higher energy density than the current battery technologies. With diesel fuel providing several times more energy per kilogram than lithium-ion batteries, UAVs can carry heavier payloads or fly longer distances without refueling or recharging. However, hybrid powertrains use a combustion engine to produce electricity, which is subsequently used to drive electric motors. Electric motors provide the benefits of combustion engines in terms of range and endurance, while generating less noise and lower emissions.

2.2. Takeoff and Landing

According to Table 3, UAVs are generally classified according to their physical structure, such as fixed and rotary wings. During flight, rotary-wing flight vehicles produce lift from spinning wings. In cruise flight, they are often limited by the efficiency of cruise speed and, therefore, have a limited range [33]. However, rotary-wing vehicles exhibit excellent hovering and vertical takeoff and landing capabilities. Along with standard helicopters, rotary-wing vehicles also adopt multicopter designs. Rotors are often assembled in stacks at the expense of aerodynamic performance to combat the large footprint caused by the large rotor surfaces [34]. Lift-fan vehicles have compact dimensions. Under hovering conditions, lift-fan vehicles are substantially less effective than rotary-wing vehicles. However, the compact dimensions of lift-fan vehicles allow them to be designed to be the same size as automobiles. As lift-fan vehicles have a small cross-sectional area, they produce high downward velocities to generate sufficient thrust, which produces a significant amount of noise. Despite the advantages of enclosed fans in ground-handling safety, reducing noise is a significant challenge for this type of vehicle. Fixed-wing flight vehicles are substantially faster and more efficient than rotary-wing vehicles during cruise flight. While extending travel distances, this structure may reduce the hovering flight features and VTOL [35]. The deployment of large rotor areas necessary for effective hovering flight is also constrained by integration issues and the desire to increase cruise efficiency [36]. Figure 4 illustrates the eVTOL horizontal and vertical landing approaches.

Table 3. Advantages and limitations of aerial vehicle types.

Category	Advantages	Limitations
Single rotor	VTOL and hover flight supported	Costly
	Long endurance and large coverage	Require more training
	High payload capability	High risk
Multi-rotor	VTOL and hover flight supported	Low area coverage
	Support both indoor and outdoor	Limited payload
	Accessibility and easy to deploy	Short flight time
	Small and cluttered areas of coverage	Low stability in wind
Fixed-wing	Long endurance and large coverage	No VTOL and hover
	High mobility	Require more training
	Great stability	High price
Hybrid	Long endurance and large coverage	Extra weight
	VTOL supported	High cruise drag forces
	Easy to take off and land	Under development

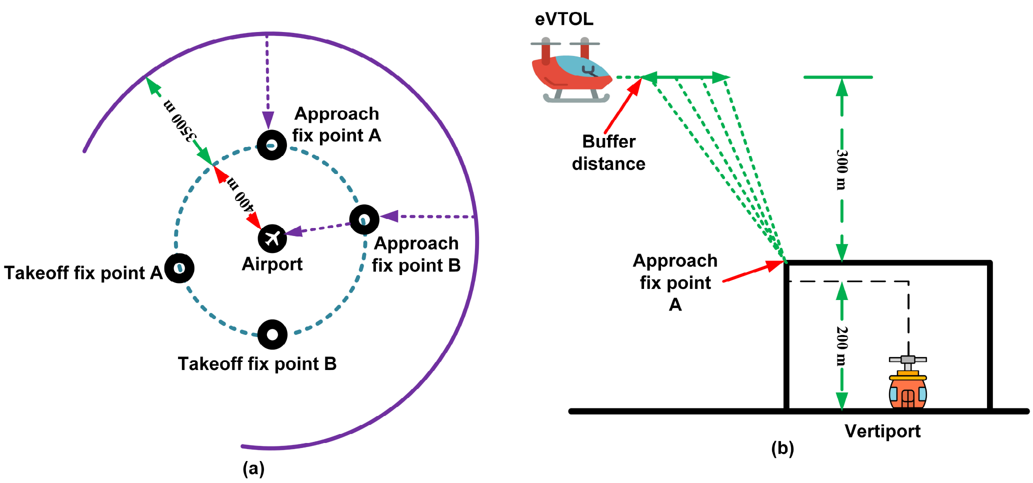

Figure 4. (**a**) Horizontal and (**b**) vertical landing approaches of eVTOL aircraft.

Lift and cruise vehicles use two distinct power tracks/trains for VTOL and flight. There is no need for tilting mechanisms because all components for thrust generation are placed such that they provide thrust in the direction opposite to that of the desired motion. This reduces system complexity and enables the design of all rotors and propellers to their maximum potential. The tilt-wing aircraft and tail-sitters employ the same propulsion techniques during all flight phases. Therefore, it is necessary to modify the propulsion system design. However, efficiency losses can be minimized by tuning specialized cruise propellers for different circumstances and turning off VTOL propellers during flight. In these systems, efficient aerodynamic and cruise flight configurations are possible by shrouding or folding unnecessary propulsors [37]. However, in a tilt-wing configuration, tilting is mandatory, resulting in a heavier and more complex system. Tail-sitter arrangements turn the entire vehicle by 90° for VTOL; in conjunction with effective fixed-wing cruising. The airplane stands straight, with its front facing towards the sky during takeoff. The vehicle gradually tilts to a horizontal flight position after takeoff until it reaches a cruise flight altitude. Although tilting mechanisms are not necessary, passenger comfort is a crucial component of tail-sitter UAM vehicles [38]. Reasonable solutions are required for a complete tilt-over to be safe and comfortable for passengers. For a comprehensive understanding of UAM vehicle types and their influence on urban aerial transportation, detailed aerial vehicular systems and different vehicle types are required. Communication techniques, navigation, geographical location, route, network, and flight scheduling are necessary to study the relationship between certain aircraft characteristics and transport systems.

2.3. Level of Autonomy

UAM aerial vehicles can be operated in many different ways, including autonomously, piloted by a human, or by a combination of both [39]. The level of autonomy and the need for a human pilot may differ depending on the type of aerial vehicle and the regulatory environment. Taking advantage of sensors and navigation systems, fully autonomous aerial vehicles can operate without the involvement of a human pilot, make independent decisions, and operate without guidance from a human. These systems can offer increased efficiency, reliability, and safety benefits. In addition, fully autonomous vehicles can operate in hazardous or complex environments that humans cannot access. However, fully autonomous vehicles also present their own challenges, such as the need for robust and reliable sensors, navigational systems, and communication systems, as well as the possibility of technical malfunctions [40]. Remote controls or other interfaces control human-piloted aerial vehicles. These systems offer the benefits of human judgment and decision-making, but they may be less efficient and less safe than fully autonomous systems, especially in complex or dynamic environments. There are also hybrid systems that combine autonomous and human-piloted operations, such as systems that allow the pilot to take control of the UAV in an emergency or when the UAV encounters unexpected situations [41]. Overall, UAV autonomy is crucial in designing and operating these systems. UAV control plays a crucial role in urban airspace development and UAV integration in traffic management.

UAM autonomy levels can be classified based on human–technology collaboration. The authors of [42] and the SESAR framework [43] classified UAM autonomy levels from level 0, indicating no automation, to level 5, signifying full automation with swarm capabilities. Gradations of levels enable an evolving UAM automation system that keeps pace with ongoing technological advancements. Level 0 indicates no automation, where the UAM fully operates under human control and pilots handle navigation, system monitoring, takeoff, and landing. Level 1 indicates assisted automation, where the pilot controls the vehicle, but automated assistance is involved, such as stability control, altitude holding, and basic navigation. Level 2 represents enhanced automation, where major UAM flight operations are performed by an autopilot but require pilot supervision. The autopilot handles flight cruising, takeoff, and landing on a predefined path. Level 3 represents a high degree of automation, where the UAM system autonomously handles all flight aspects, such as navigation, obstacle avoidance, and real-time decision-making during flight. Level 4 is fully automated, where the UAM can operate autonomously without a pilot during flight. Level 5 is swarm automation of UAM, in which UAM can be deployed on a large scale. Using swarm automation, UAM can operate autonomously and self-organize with advanced artificial intelligence and machine learning to adapt to various scenarios.

2.4. Integration with Existing Networks

The integration of UAM systems into existing transportation networks is an important consideration. UAM systems, such as urban air taxis and delivery drones, have the potential to revolutionize human transportation and goods delivery [44]. Compared with traditional modes of transportation, UAM provides a convenient and efficient alternative. UAM systems may be integrated into existing transportation networks to provide users with a seamless and convenient travel experience. Public transit or ride-sharing services [45] can allow users to easily book a UAM ride or delivery through a single platform or app. They can minimize the need for additional infrastructure, such as dedicated landing zones or charging stations. Integrating UAM systems with existing transportation networks can also reduce ground congestion and improve air quality by reducing the number of vehicles on the road.

2.5. ATM

UAM poses several challenges that cannot be addressed using conventional ATM techniques [46]. All types of aircraft, including commercial and military flying vehicles, are controlled by ATM services, both in current and next-generation ATM systems. Unlike commercial flights, which are intended to ensure the safe transportation of passengers and containers between well-known airports, military planes protect the country's airspace from enemy attack and support ground troops. Increasingly, UAM will require a separate ATM system or an adapted system that can accommodate on-demand, high-volume, and short-range flights near urban areas [47]. Because these autonomous aircraft will provide their services in densely populated urban airspaces, they are expected to operate under tighter and more stringent rules than the present ATM system. Any UAM traffic management system must also handle unusual occurrences safely without jeopardizing overall performance [48]. A comparison of traditional ATM with UAM ATM is shown in Figure 5. The UAM ATM framework requires collaboration between air and ground collaboration systems, which combines the planning of urban airspace, traffic rules, and the management of ground facilities.

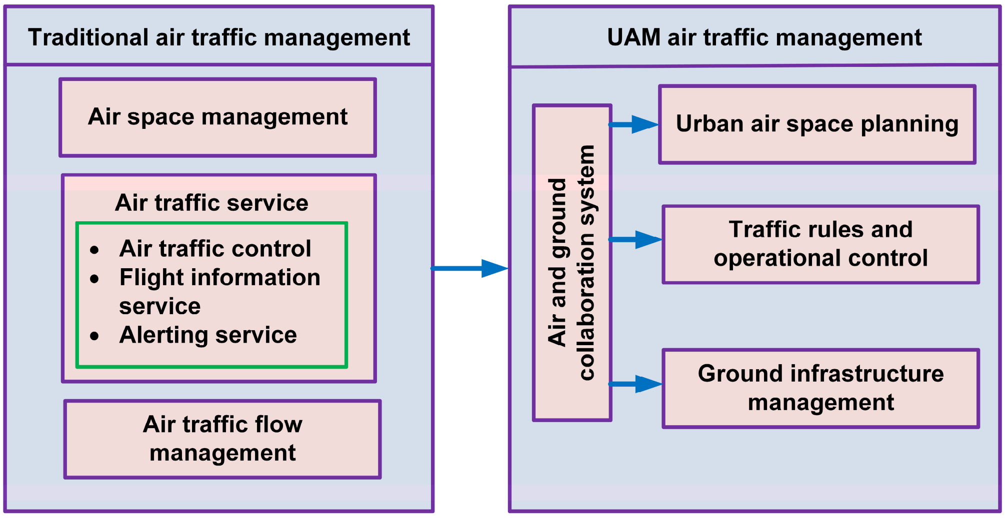

Figure 5. Comparison between traditional ATM and UAM ATM.

Unmanned aerial aircraft (UAS) traffic management is a proposed traffic management system for tiny UAS that adopt a federated approach to maintaining airspace access [49]. Various safety-oriented services, including aircraft separation [50] and geofencing [51], are widely utilized in ATM. However, when using this strategy, there may be problems with scalability, both in terms of the size of the UAM vehicles and the density of UAM traffic. There is an increasing need to investigate the architecture of a well-developed ATM system capable of securely and effectively controlling UAM traffic. Figure 6 shows the basic framework of UAM ATM. UAM traffic management issues for eVTOL vehicles are divided into three categories: airspace planning, traffic rules and operational control, and ground infrastructure management [52].

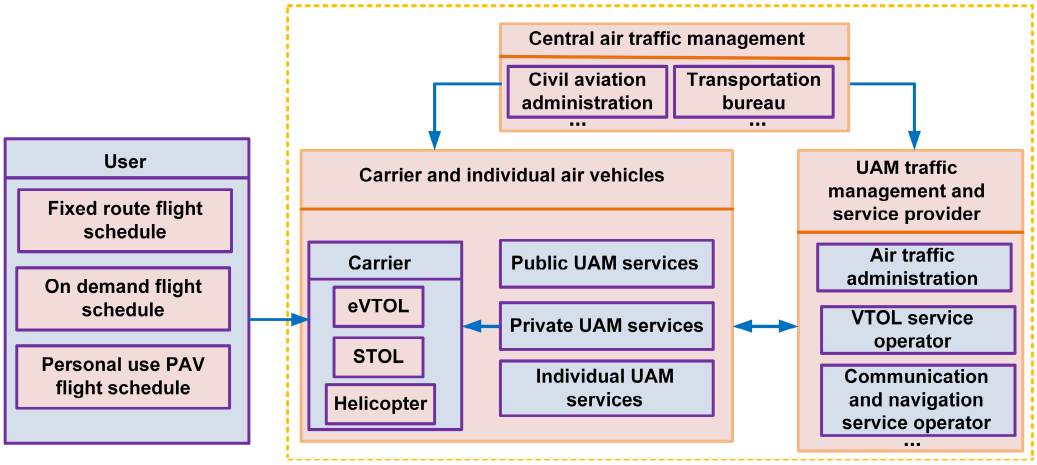

Figure 6. UAM ATM framework.

UAM needs to refine the airspace planning and management of existing low-altitude uncontrolled airspaces. A flight-level airspace structure can provide a better balance between operational safety and airspace capacity. To plan ground infrastructure, urban origin–destination data can be used to predict the traffic demand of UAM. The management of vertiports will have a significant impact on the air route network as well as the layout of communication, navigation, and monitoring stations. UAM will be faced with more complex integrated operation scenarios involving manned and unmanned vehicles in terms of traffic rules and operational control. UAM traffic rules must be innovated and remain compatible with existing air traffic regulations [53]. A high-bandwidth communication system will allow UAM operational control to move in the direction of autonomous decision-making and self-piloting based on air–ground coordinates.

Furthermore, the SESAR automation roadmap [43] provides a vision for the future of ATM and identifies key phases for integrating automation technologies, particularly in light of the increasing complexity of airspace usage, as well as the integration of new aircraft types into UAM, including unmanned aerial vehicles and air taxis. The roadmap focuses on improving the safety, efficiency, and capacity of air traffic systems through automation. According to SESAR, there are different levels of automation across various aspects of ATM, ranging from flight planning and departure sequencing to conflict detection and resolution, in-flight automation, and landing coordination.

Currently, both the FAA and EASA are heavily investing in the integration of UAM and UAVs into existing ATM systems. Their work focuses on developing air traffic control systems (e.g., UTM in the U.S. and U-Space in Europe), airspace integration, safety standards, and certification processes for new aircraft types, such as eVTOL aircraft. Key efforts include ensuring the safe use of airspace, support for multi-vehicle operations, and the development of regulatory frameworks to accommodate the growing volume of low-altitude aviation in urban areas. A new airspace management system is being developed by the FAA to support UAM operations, which require coordination between manned and unmanned vehicles, particularly at low altitudes in urban areas. An important aspect of this process is the development of UAM corridors and vertiports (takeoff and landing zones). As part of its studies on the integration of UAM vehicles into the national airspace, the FAA will focus on how eVTOL aircraft and drones interact with traditional aircraft. Both traffic management and collision avoidance technologies are used to ensure that UAM operations do not interfere with existing air traffic. The U-Space concept, which is similar to the FAA's UTM initiative, is one of the EASA's key initiatives for the integration of UAM into the European airspace. It consists of a set of services and procedures that facilitate the safe integration of drones and electric VTOL aircraft into the European airspace. This framework supports automated flight management such as conflict detection and resolution, real-time flight tracking, and dynamic airspace management, which are essential for managing the high volume of UAM vehicles in cities.

2.6. Energy Efficiency

Battery endurance is a significant challenge for electric UAVs and drones because it imposes severe constraints on their operational time. There are several solutions to improve the endurance of electric UAVs, such as more efficient rotor configurations, lightweight materials, and trajectory planning to minimize energy consumption [54]. One of the most promising approaches to improving electric UAV endurance is carefully planning their trajectories to minimize energy consumption. This approach can involve flying at lower altitudes, taking shallower descents, and avoiding obstacles or other sources of drag. Studies have shown that energy efficiency drops with increased cruise altitude, so flying at lower altitudes can improve electric UAV endurance [55]. However, it is imperative to note that optimizing a single factor, such as energy consumption, may result in a suboptimal solution when other factors, such as safety and community acceptance, are also considered. Therefore, the design of airspace structures and routes for UAVs should consider the trade-offs among energy consumption, efficiency, safety, and community acceptance [56]. The impact of UAM vehicle type on urban aerial transportation requires a thorough assessment of aerial vehicular systems and different types of vehicles [57]. Generally, all aviation modes that permit individual or personal mobility are defined as personal aerial vehicles (PAVs). Some documents refer to PAVs as flying cars or roadable aircrafts. NASA first explicitly mentioned PAVs within its NASA/LANGLEY projects relating to personal air vehicles (PAVs) and dual air/road transportation systems (DARTSs) [57]. These vehicles aim to provide a breakthrough in personal air mobility by reducing travel times and increasing reach, thereby enhancing the quality of life.

3. Communication Techniques in UAM

Effectively managing multiple aircraft movements in a complex urban environment is a key challenge in UAM. To address this challenge, it is crucial to develop communication systems that can facilitate the exchange of information between UAM vehicles, air traffic control, and ground-based infrastructure, as shown in Figure 7. These systems must be capable of managing high data volumes, supporting real-time communication, and remaining resilient to interference and disruptions. A successful UAM solution must take into account the challenges and differences between various environments, such as water bodies, rural locations, and urban centers. Line-of-sight (LOS), non-line-of-sight (NLOS), and blind-line-of-sight (BLOS) links, which experience sporadic obstructions, pose a greater threat than the current aviation environment [58]. Furthermore, accuracy and latency are key in navigation signaling. The recent surge in interest in air taxis, driven by ride-sharing services like UBER and Lyft, underscores the need for system robustness, safety, and connectivity in realizing such services [59].

UAM communication systems must be extremely reliable, secure, accessible, and satisfy high data integrity standards. While UAM systems exist independently of current communication systems, commercial objectives require low-cost, lightweight, and power-efficient customization. Researchers have attempted to improve flight safety and security with the help of machine learning and artificial intelligence. Services hosted in the cloud are being studied to ease the cargo delivery process, such as various types of communication links for essential telemetry data and connectivity, obstacle avoidance systems, and related sensors [60]. Other communication systems include flight control, propulsion, quick navigation, surveillance, terrain mapping, and emergency landing systems.

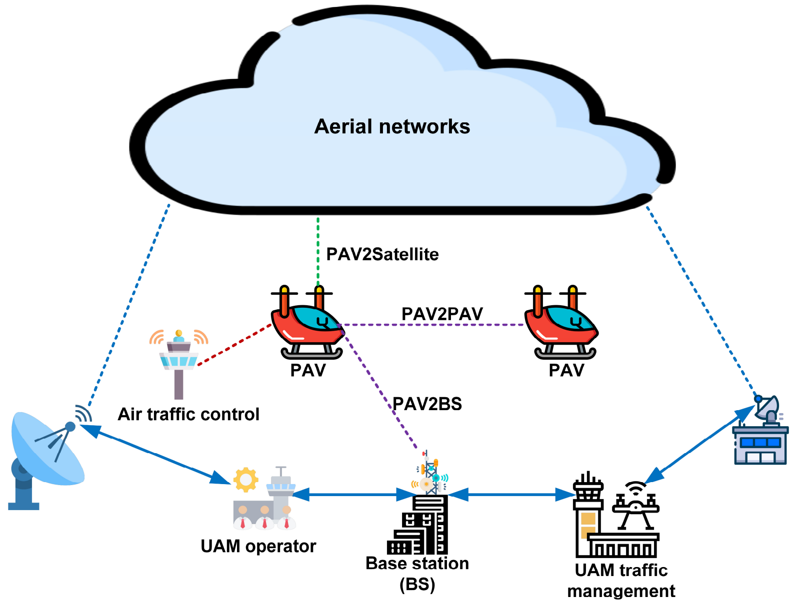

Figure 7. UAM communication and information flows.

3.1. Data Type

A key aspect of the operation of eVTOLs is their connection to communication networks. To achieve high data rates, communication should be ultra-reliable, secure, and low in latency. In addition, passengers should be able to access voice and data networks with a stable connection that provides satisfactory quality of service (QoS) [61]. The autonomy of eVTOL communication is another aspect that contributes to its reliability and robustness. UAM vehicles have two types of data: command and control (C2) and non-C2. As shown in Figure 8, C2 contains information related to flight control, safety systems, navigation, and communication. C2 links ensure the safe operation of an eVTOL aircraft when it is unmanned [62].

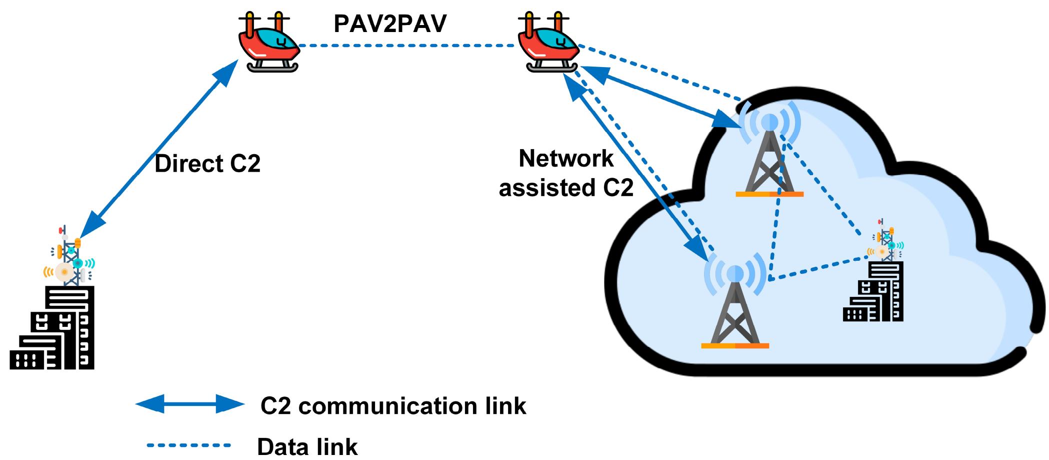

Figure 8. C2 link in UAM communication.

This data type is constantly changing because the real-time evaluation of flight operations for potential aborts will require a significant amount of data from flight system diagnostics. The handling and operation of aerial vehicles may benefit from live visuals, as they facilitate remote operations, for example, through artificial intelligence applications across cloud services. The non-C2 category includes all additional data such as post-flight data or passenger information [63]. System design and requirements must consider these difficulties to offer such a connection for control and non-payload communication (CNPC).

The capabilities of the CNPC system are also extended to handle various informational types, including command messages, target track data, non-payload data, air traffic control (ATC) voice relay navigation aids, non-payload video downlink (DL) data, air traffic data relay, and airborne weather radar DL data. The C2 communication connection should also support secure communication between the aerial vehicle and the control center to provide secure and dependable UAV flight operations [64].

3.2. Spectrum and Carrier Frequencies

With high-density UAM, the number of devices accessing the spectrum is predicted to increase significantly in the near future. Furthermore, the RF spectrum capacity approaches its limits, further exacerbating the issue of spectrum scarcity [65]. Owing to the limited suitability of existing orthogonal multipoint access schemes for future eVTOL networks, several studies have been conducted on the coexistence of terrestrial and aerial devices such as UAVs in three-dimensional environments, which can be easily generalized to UAM. There are two promising multiple-access schemes for UAM communication: non-orthogonal multiple access (NOMA) and rate-splitting multiple access (RSMA) [66]. Communication systems experience congestion in their bandwidth and transmission spectrum. For this reason, UAM communication systems must also utilize modern frequency bands such as sub-6 GHz, 3GPP bands, millimeter wave (mmWave) bands (24–86 GHz), and aviation and low-Earth orbit (LEO) bands. Owing to open space and extremely high tropospheric attenuation, the radio signal range must be reduced to increase the frequency. The highest bandwidth efficiency and data rates are offered by 5G mmWave systems. However, they are mostly LOS and have short ranges, making them suitable only for metropolitan areas. UAM systems must use low data rates in both rural and outlying areas.

3.3. Reliable Communications

Due to their low altitude, aerial vehicles and standard aviation systems, such as helicopters, differ significantly in their proximity to structures and barriers on the ground. This difference creates a communication challenge known as shadowing and obstruction [67]. Because of these effects, a connection may be attenuated to the point of failure. As a result, several links are required to connect the aircraft to command-and-control stations. Since air-to-ground communications are more power-efficient and less time-consuming than satellites and HAPs, they are generally preferred for LOS links [68]. BLOS refers to a link that can only be accessed using methods other than a ground station (GS), such as other aerial vehicles, platforms, or satellites. The NLOS scenario is often the most difficult in terms of power and channel degradation because of multipath components in the environment. Although NLOS is an uncommon scenario in UAM, it should be considered once the UAM is completely distributed. In short, the control and command connections should be incredibly robust in the face of interference and should recover quickly after a breakdown. eVTOL aircraft face a number of challenges in terms of connectivity because of the highly dynamic UAM environment and the large number of obstacles present in cities. For continuous connectivity, the technology must be able to achieve BLOS communication. Reliable coverage ensures the continuous flow of information needed for navigation and surveillance [69], paramount for safe operations. Several opportunities exist for high-coverage and high-reliability communication, such as next-generation cellular networks, aerial platforms, satellites, RF, and reconfigurable intelligent surfaces (RISs).

A comprehensive analysis of the communication technologies used by UAM is presented in Table 4. In the table, we provide an explanation of the application scenario for each communication technology as well as the advantages and limitations associated with that technology.

Table 4. Comparison of communication technologies for UAM.

Category	Communication Technology	Link Types	Applications	Advantages	Limitations
Ad hoc networks	FANET	Aerial-to-aerial (A2A) and A2G	FANET-based UAM for communication and collision avoidance	Used for inter-vehicle communication and localization	Risk of link disconnection
Cellular networks	Long-term evolution (LTE)	UAV-to-infrastructure (U2I) and A2G	Surveillance and awareness messages	Low-cost deployment and large area coverage	Inter-cell interference problem and not suitable for high density of aerial vehicles
	4G	A2G and C2	Surveillance and communication between pilot and unmanned aircraft system traffic management (UTM)	Low-cost, reliable, and allows connectivity for beyond visual line of sight (BVLOS) situations	Link availability problem in rural areas
	5G	A2G and C2	Suitable for the large-scale UAM operation including control, tracking, and communication	Low latency, wide area coverage, limitless connectivity, and multi-access edge computing-enabled services	Lack of short-distance connections and limited coverage
	6G	U2I, A2A, and A2G	Monitoring, surveillance, and emergency management	High-precision localization and wide area coverage	Still not tested in real environment
Wireless local area network (WLAN)	Wi-Fi (IEEE 802.11 series and 2.4 and 5 GHz frequency brand)	A2A and A2G	Real-time information exchange between aerial vehicles	Cost-effective, high availability, high bandwidth, and high hardware availability	Limited range, interference, and high energy consumption
	X-Bee (IEEE 802.15.4 and 2.4 GHz frequency band)	A2G	Surveillance data transmission to the ground station	Longer distance compared to Wi-Fi	Limited data rate
Satellite networks	LEO and geostationary equatorial orbit (GEO)	A2G	Useful for BVLOS situations	Real-time reliable connectivity, high area coverage, robust communication, and high capacity	Low data rate, not suitable for delay-sensitive applications, and high complexity
Other networks	Long range (LoRa)	A2G and A2A	Pee-to-peer communication and surveillance	Mid-level transmission range with high throughput	Packet loss happened when transmission distance crossed more than 1 Km
	Automatic packet reporting system (APRS)	A2G	Transmission of surveillance data	Supports higher altitudes compared to 4G and 5G	Low data rate and congestion may happen in high-density aerial environments
	Automatic dependent surveillance—broadcast (ADS-B)	A2G	Surveillance and tracking	Acceptable range and accuracy	Security concern

3.4. Data Rate and Latency

To facilitate rapid data transmission, eVTOL aircraft require high data rates and low-latency communication links. UAM communications have varying data rate requirements depending on the application. Low data rates can be used for voice applications and simple C2 messages. Remote pilot operation (RPO) via video streaming and autonomous technology has more stringent requirements, exceeding 100 Mb/s [70]. Moreover, high rates are expected for passenger infotainment systems. City bandwidth shortages are the primary obstacle to enabling high data rates. Different frequency ranges, such as mmWave and terahertz (THz), can be a good solution for achieving high-throughput communications. UAM requires low latency demands due to eVTOL operations in crowded cities. To enable RPO and autonomous navigation, latency as low as 10 ms is required [71]. Reducing communication distances and eliminating relays is the key to achieving low-latency communication. A low-latency connection can be achieved by decentralizing the communication using sidelink or device-to-device (D2D) technology. In addition, eVTOL vehicles can fly at very high speeds (300 km/h) and may be partially or entirely autonomous. The position of an eVTOL must be continuously shared while receiving the positions of other eVTOLs and other flying platforms, such as drones [72]. Moreover, they should be continually informed of air traffic, weather, and unexpected occurrences. Thus, a new air interface technology is required to allow eVTOLs to operate at the required data rate and latency.

3.5. Communication Zone

Future aviation communication systems will require links with higher throughput and longer communication distances, which opens up future study areas. Compared with terrestrial communications, power and spectral efficiency are not important in UAM settings [73]. To cover long distances, aerial communication regions are often larger than those of terrestrial communication systems. Since all of the power must be conserved in aviation stations, as it needs to be distributed among numerous power-consuming systems, aviation channels are frequently modeled as Rician fading channels, where the Rice factor, also known as the K-factor, can range from simple additive white Gaussian noise (AWGN) to a more complex and distortive Rayleigh fading channel [74].

3.6. Secure Communications

UAM safety and security are among the main concerns in the design of communication protocols. Because eVTOL aircraft are most likely to operate in urban areas and among pedestrian traffic, they must be safe not only for people on the ground but also for pilots and passengers. Secure communication is crucial for UAM operations to prevent the hacking and jamming of eVTOL aircraft. Malicious hackers can cause disastrous damage if they gain control over one or more eVTOL aircraft. These consequences could be fatal for pedestrians, eVTOL vehicles, passengers, and buildings. Various recent technologies, such as blockchains, machine-learning security algorithms, and quantum computing, can be used to secure communication [75].

3.7. Network Topology

Multipoint receiver and transmitter schemes can be employed in UAM networks using centralized or decentralized 5G techniques. Multiple links can also be offered by FANETs, which often require more sophisticated network-level techniques such as adaptive routing [76]. The use of high-altitude platforms and satellite communications provides several connectivity options. C2 connections are expected to have low data rates and bandwidth due to the type of communication, such as 300 kbps for compressed videos [77]. On the other hand, non-C2 is often used for complex applications and high-data-rate requirements. In UAM vehicles, passengers require a throughput equivalent to that of terrestrial networks. UAM vehicles communicate at 300–600 m via an aerial network hierarchy that includes geostationary and low-orbit satellites. The altitude of satellites is much higher than that of HAPs, varying from a few hundred kilometers for LEO to a few thousand kilometers for GEO. In contrast to HAPs, satellites are located at higher altitudes and their path lengths are significantly longer. Moreover, the satellite-to-eVTOL link experiences additional losses related to atmospheric loss, antenna misalignment, and polarization [78]. Furthermore, the altitude of the satellite and the angle between the satellite and the eVTOL also play a critical role in the design of the satellite-to-eVTOL link. Figure 9 illustrates multi-layered network topology management and communication. Future UAM networks should be able to adapt to changes in the network topology, environmental dynamics, and evolving user requirements. Software-defined networks (SDNs) are an effective strategy for coping with rapid topology changes caused by UAM [79].

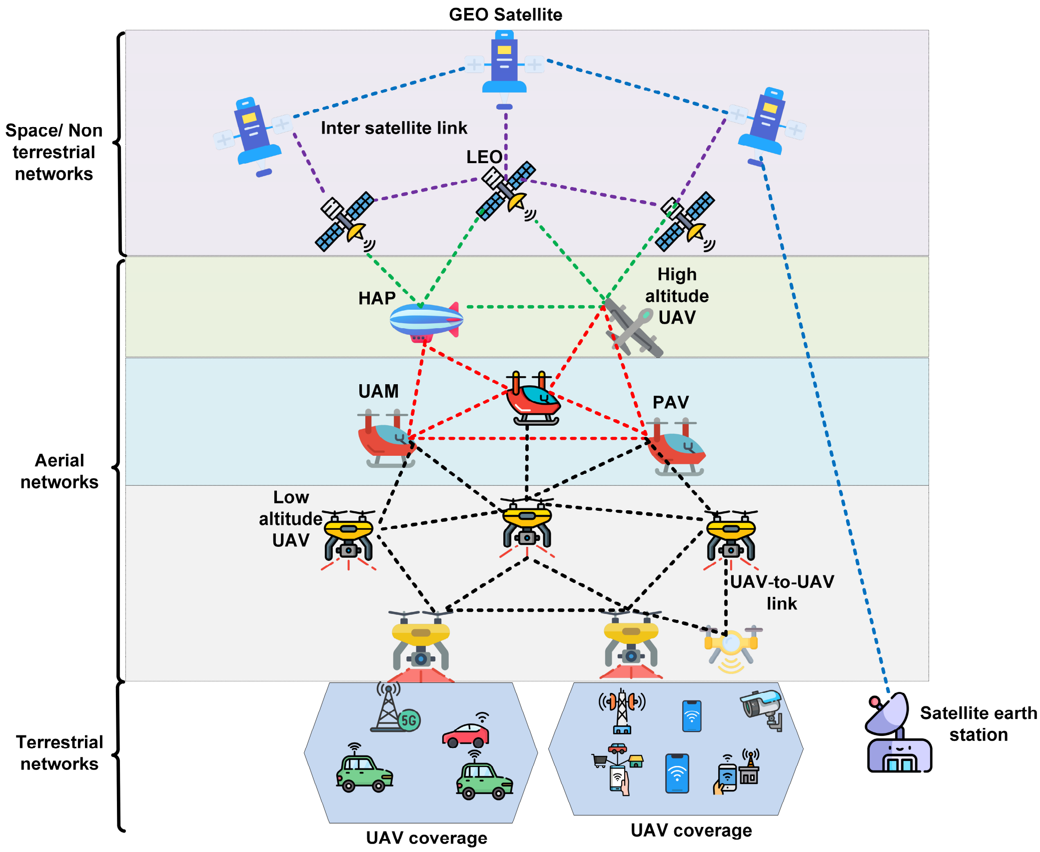

Figure 9. Multi-layer aerial communication networks.

3.8. Navigation and Localization

With an increase in the number of UAM nodes, a high-accuracy positioning system is necessary for situational awareness and self-awareness. UAM operations require robust tracking algorithms and fusion sensors, which provide accurate information and positional awareness. The authors of [80] pointed out that current global navigation satellite systems (GNSSs), including GPS, Galileo, the quasi-zenith satellite system (QZSS), and the global navigation satellite system (GLONASS), can be improved to improve the accuracy of UAM navigation systems. Since GNSSs were created for LOS situations, they are better suited to moderately populated metropolitan regions. According to [81], the combination of all the stated technologies will provide multiband, multimode receivers with high accuracy, but their price is too high for low-cost commercial applications.

For challenging environments devoid of LOS, the authors of [82] use 5G new radio (NR) at frequencies ranging from 450 to 6000 MHz and 24.25–52.6 GHz to investigate a hybrid alternative to GNSS. To achieve positional awareness with high precision even in richly distributed NLOS fading channels, the authors of [83] assessed six carrierless mmWave waveform types using impulse radio (IR) and deduced a comparison of range accuracy with multiple carriers. In the experiment, they employed an energy detector with a threshold that has low computing complexity and offers decent range accuracy rather than outstanding accuracy, such as provided by a correlation detector. The authors discussed these waveform options for 5G NR cellular network D2D communications and vehicle location. As discussed in [84], none of the systems suggested any implementation that ensures 99.99% uptime anywhere under any condition; therefore, hybrid techniques are considered the most effective navigation solution.

Communication, navigation, and surveillance (CNS) facilities on the ground are important components of UAM. CNS provides technical support for UAM traffic operations. Compared with transport aviation, UAM requires vehicles to operate at a higher traffic density. The emergence of 5G communication technology may be crucial for solving CNS problems, as indicated in Table 5. There is insufficient airspace coverage from the base stations on the ground. UAM transport requires improvements in the selection of ground base stations and antenna layout (such as adjusting antenna radiation direction, transmission power, and signal gain) to achieve comprehensive coverage of urban low-altitude airspace. Table 5 compares CNS requirements for traditional air transportation and UAM. Various sensors and techniques can be used for localization, often in combination, to improve accuracy and robustness. A few common technologies used for localization are inertial measurement units (IMUs), light detection and ranging (LIDAR) systems, and GPS. Some commonly used localization techniques are visual odometry (VO), simultaneous localization and mapping (SLAM), beacon-based localization, RF localization, particle filters, and Kalman filters [85]. There is a significant difference between VO and SLAM in that VO focuses primarily on local consistency and aims to incrementally estimate the path of the camera/UAV position after a change in position, as well as possibly optimize local parameters. However, SLAM aims to achieve a globally consistent estimate of the camera/UAV trajectory.

Table 5. Comparison of CNS requirements for traditional air transportation and UAM.

Service	Traditional Air Transportation	UAM
Communication	Aircraft communication addressing and reporting system, very high frequency (VHF) ground-to-air data communications link.	Ground base stations provide 5G communication networks and onboard ad hoc network communication.
Navigation	Very high-frequency omni-range, distance measuring equipment, and non-directional beacons are used for navigation purposes. The navigation station provides radio navigation, and the instrument landing system provides landing guidance. GNSS and IMU provide combined navigation systems.	GNSS and IMU-integrated navigation, urban domain space high-resolution three-dimensional (3D) map service, location-based routing services, and online visual recognition-aided positioning are used for navigation.
Monitoring	There are three types of radar surveillance: primary radar, secondary radar, and automatic dependent surveillance–broadcast technology.	Ground-based 5G communication links, centralized cooperation related to monitoring, and FANET communication.
Regulatory	In the radar control station, the process is directed by the controller sequence according to the route, approach, airport, and area.	Autonomous flight ground control, centralized control scheduling, and manual responsibility for emergency response and surveillance.
Flight information	Provide air traffic service, automatic information broadcast service—communication information includes the airport and route meteorological conditions.	Airspace traffic information, vertical takeoffs and landings, airport slot status, detailed meteorological information for urban and rural areas, and route stream information.
Emergency system	When an aircraft is in an emergency situation, the control units alert the airspace service, based on a secondary-mode answering machine code warning system.	Advertising throughout all airspace via the 5G network, on-board equipment takes action when a fault is detected.

3.9. Collision Detect-and-Avoid (CDAA) Systems

Utilizing terrestrial CDAA for typical UAS terminal area operations is required to integrate UAS into urban and near-urban airspaces. The ground station should know all planned flights and their paths as well as the current airborne flight locations and flight paths. Aerial vehicles should also be able to communicate with the GS and relay the coordinates of other UAS in the operating vicinity. Advanced driver assistance systems (ADASs) with forward-collision warning tools are used in the context of autonomous ground vehicles. These systems use sensors such as radar, cameras, laser imaging, and LIDAR to enable semi-autonomous driving in autonomous mode [86]. In the aviation industry, aircraft collision avoidance systems are commonly referred to as ACAS or TCAS [87]. An ACAS is used in aviation to prevent mid-air collisions between aircraft. It provides real-time warnings and instructions to pilots to avoid collisions with other aircraft. TCAS is a specific subset of ACAS. In commercial aviation, this system monitors the airspace around the aircraft and alerts pilots to potential traffic conflicts, which are followed by maneuvering instructions to avoid a collision.

Frequency-modulated continuous-wave (FMCW) radars are employed in CDAA systems on ground vehicles and may be implemented in UAM. The authors of [88] studied the ghost target scenario caused by FMCW radar-to-radar interference. ADASs with forward collision warning tools are used in new autonomous vehicles to reduce interference. This system uses sensors, such as radar, cameras, and LIDAR, to enable semi-autonomous driving in automobiles. To reduce interference, the authors of [89] suggested altering the chirp slope in the time–frequency domain (DTF). These systems are appropriate in situations with a high likelihood of a UAV-to-UAV collision, the avoidance of which is made possible by cutting-edge interference-mitigation methods. With the goals of standard compliance and dual-radar and communication functions for UAM, chirp signals are created for DFT-s-orthogonal frequency division multiplexing (OFDM) as frequency–domain spectral shaping using Bessel functions and Fresnel integrals.

The authors of [90] propose a Markov decision process (MDP)-based method called FastMDP for collision avoidance in urban air mobility. The authors demonstrated that FastMDP outperformed other collision avoidance algorithms in terms of safety and computational efficiency, even when the aircraft had imperfect knowledge of each other's intentions. In the FastMDP, the initial aircraft state and the limits that must be maintained are the states, and the aircraft defines the corresponding action. A reward is defined by the reachable states for each action over the time horizon to address the challenge of safely managing multiple eVTOL aircraft in congested urban areas for efficient and on-demand air transportation. In [91], the authors proposed a computational guidance algorithm with collision avoidance capability, formulated as an MDP, and solved it using a Monte Carlo tree search algorithm. Their experimental results show that the proposed algorithm performs better than the optimal reciprocal collision avoidance strategy in reducing conflicts and mid-air collisions.

3.10. Routing

UAM vehicles move at very high speeds, causing frequent network topology changes, as illustrated in Figure 10, which can make routing a challenging task because previously available routes may no longer be valid. To address this challenge, researchers have proposed several routing algorithms and protocols specifically designed for UAM networks. Recently proposed algorithms and protocols consider the dynamic nature of the network and aim to find the most efficient route from one node to another [92]. Since technology has advanced and GPS locations can be tracked, several location-aware dynamic routing algorithms have been proposed in recent years. By using location-aware routing models, real-time solutions have been developed, including aerial vehicle operation, vehicle location, vehicle charging and maintenance requirements, and customer characteristics (e.g., pick-up/drop-off location, pickup time, willingness to rideshare, and number of passengers). Moreover, high-performance computing is required to supervise UAM operations and provide navigation services to eVTOL vehicles [93]. Utilizing modern networking technologies, such as cloud/fog computing, edge computing, and digital twins, UAM can access large amounts of computing power without onboarding expensive hardware.

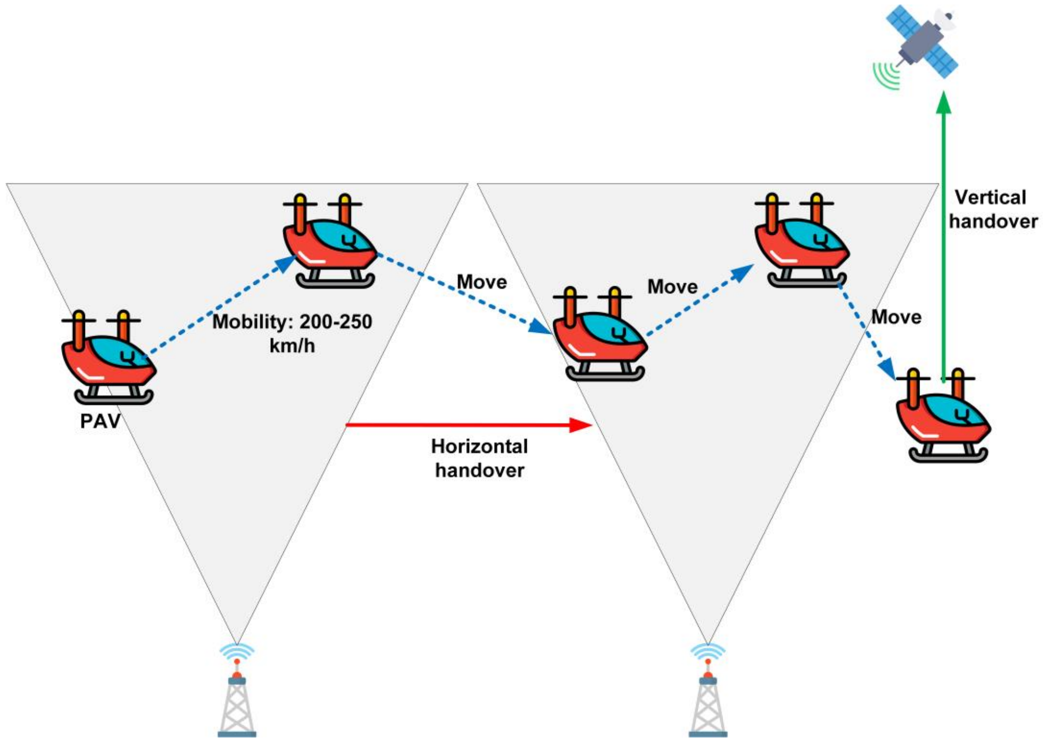

Figure 10. Mobility effect on UAM communication.

3.11. Path Planning and Mission Planning

In path planning, an algorithm determines the path or trajectory that a flying vehicle should follow from its current position to a specific goal or destination while avoiding obstacles and following certain parameters. A path planning algorithm considers the dynamics of the vehicle; the environment, including obstacles; the terrain; and other relevant factors to determine the most efficient and collision-free route. Path planning aims to determine a feasible and safe trajectory that guides the vehicle from beginning to end, while optimizing parameters such as time, energy consumption, and smoothness. Three main algorithms are used in UAM path planning: graph-searching algorithms, sampling-based search algorithms, and meta-heuristic algorithms [94,95]. The first are foundational algorithms such as Dijkstra and A* algorithms. The second category includes probabilistic algorithms, such as the probabilistic roadmap method (PRM), and rapidly exploring random trees (RRTs). The third category comprises algorithms derived from natural or bioinspired algorithms, such as evolutionary and swarm intelligence.

Alternatively, mission planning involves defining and organizing a set of tasks or objectives that must be accomplished for an autonomous system to accomplish a specific objective. To accomplish its mission successfully, the flying vehicle must identify the sequence of actions, waypoints, and goals that need to be followed [96]. A mission plan consists of multiple aspects, including planning for individual segments, allocating tasks, managing resources, and coordinating multiple vehicles or agents, as necessary. It optimizes the allocation of resources and tasks to achieve the end goal of the mission in the shortest possible time. Figure 11 illustrates the various factors that are considered in mission planning.

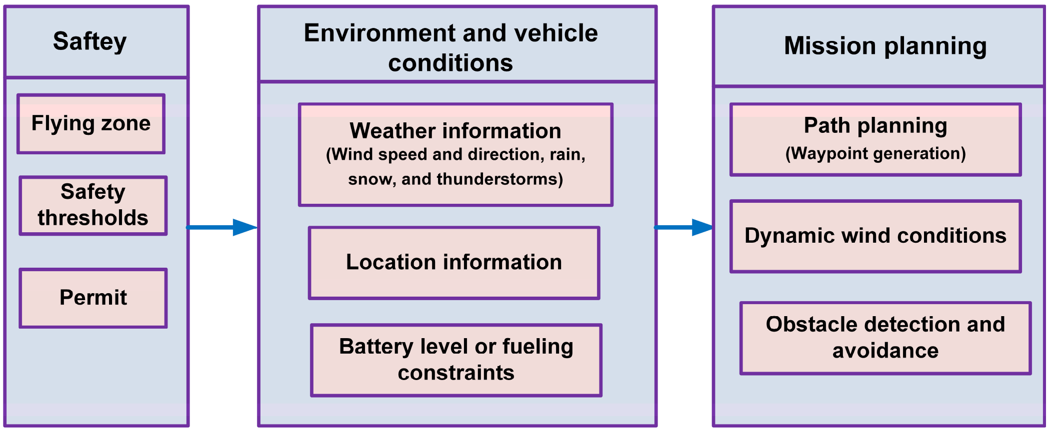

Figure 11. Overview of UAM mission planning.

3.12. *Flight Scheduling*

Unlike traditional aircraft, eVTOLs are constrained by their battery life, which limits their flight time and makes it difficult to preschedule their arrivals. To address this challenge, researchers have proposed several methods for scheduling eVTOL arrivals at vertiports, equivalent to an airport for UAM. Vertiport has facilities for customer pickup or drop-off, charging, maintenance, and several docking stations for aerial vehicles [97]. Scheduling methods include position shifting, dynamic programming, branch-and-bound, branch-and-price, and data splitting, as well as various heuristic algorithms [98]. However, these methods are not specifically designed to consider the eVTOL constraints. These constraints include limited battery life, airport capacity, and remaining fuel. Therefore, there is a need for further research on this topic to develop models and algorithms tailored explicitly to the scheduling of eVTOL arrivals at vertiports.

4. Opportunities and Challenges

Although the UAM mission is to provide safe, sustainable, economical, and accessible transportation, it faces obstacles such as public acceptance and public safety. The deployment of UAM can be made easier if there are several lucrative operating criteria, including flying limits over residential areas, nighttime operations, inclement weather, and the development of green technologies. Furthermore, to build infrastructure and scale operations, both the private sector and the government must cooperate and invest significantly in the development of VTOL vehicles and UAM. Future research, strategic planning and implementation, and an analysis of UAM implications are necessary to balance economic interests, technological progress, and the public good.

The safety of aircraft and humans must be ensured during UAM operations. Despite the considerable advantages of UAM, there are also a number of limitations, such as the difficulty of maintaining safe flight at low altitudes, the need to avoid hazards such as collisions, and the effects of weather. The UAM flight trajectory is also significantly complicated in urban areas due to the presence of high-rise buildings. As a result of these factors, UAM implementation is very challenging in urban areas. Therefore, collision avoidance in UAM may be an exciting area of future research [99]. There are several challenges and obstacles to UAM's development, including those related to safety and air traffic control regulations, noise, public acceptance, weather conditions, environmental impacts, infrastructure, and security [100,101]. With the help of air traffic services (ATS), future research could focus on developing dynamic routing algorithms that balance customer waiting times, vehicle idle times, and travel costs. Table 6 addresses many possible roadblocks that could prevent widespread UAM adoption, and some solutions to overcome them. Several ongoing research projects in collaboration with regulatory bodies focus on airspace integration [102]. There have been limited advancements in infrastructure, thorough city integration analyses, and economic evaluations. Operational and infrastructural development requirements are also major concerns in the implementation of UAM.

Table 6. The challenges of a UAM system and potential mitigation strategies.

Challenges	Threats	Probable Mitigation Strategy
ATM	Ensure safe flight, takeoff, and landing with other aerial vehicles. Low-level and uncontrolled air space.	Federal aviation act (FAA) for ATM regulation and policy. Concept of operations guidelines by FAA.
Environmental impacts	Weather hazards pose safety risks and sensitivity to passengers and vehicles. Decrease in vehicle size poses a significant threat to aircraft.	Vehicles with different capabilities suited for a variety of weather conditions and climates. Safety procedures and emergency safety protocols. Shifting trips from UAM to other modes during travel disruption due to weather hazards.
Infrastructure	Need for stations for a network of vertiports; charging or fueling; communications, navigation, surveillance, and information technology infrastructure.	Storage of batteries in the ground to manage the power grid. Different facilities based on demand and density. Water resources for water landing aircraft and reducing development of ground infrastructure.
Privacy	Non-consensual photos, videos, or surveillance of people on the ground Privacy of user trip data.	Legislating for regulations and standards on data handling and UAM services protecting consumer data.
Safety	Personal safety. Operational safety. Cyber and physical threats such as sabotage and attack.	Passenger rating system and background checks. Individual compartments and emergency response protocols for pre-, mid-, and post-flight incidents. Develop strategies and standards for communication, navigation, and surveillance to provide system redundancy.
Noise	Noise from aircraft and scale operations.	Flight operations at high altitudes. Time restrictions and path deviations to avoid sensitive areas such as schools and hospitals.

4.1. ATM System

A critical component of UTM and ATM is the exchange of information. However, ATC, ATM, and UTM are unable to exchange geofencing information because there are no common standards or protocols. As a result, research efforts should be devoted to identifying a method for facilitating the exchange of critical information between the UTM and ATM. UTM can be implemented in each geographical region, where these UTM systems communicate and exchange information in various scenarios, including when an aerial vehicle moves between regions to accomplish its goals. Due to the lack of compatible communication protocols and compliance with common standards among these UTM systems, interoperability poses a significant problem.

4.2. Air Space Management

Centralized systems with centralized decision-making are commonly used in UTM architectures. However, a single point of failure can pose a performance challenge for centralized systems. Aerial vehicles can experience abrupt and rapid changes in their behavior and motion due to a variety of factors including weather conditions, technical malfunctions, pilot errors, and unexpected events. A decentralized system can be used to overcome the limitations of centralized-based UTM. Decentralized UTM systems face several challenges including safety, interoperability, conflict management, and response times. Additionally, if communication between aerial vehicles is lost or disrupted, there is a risk of collisions. Considering this situation, it is important to determine whether a UTM system should be decentralized, hybrid, centralized, or completely distributed. To evaluate and compare these UAM concepts, focusing initially on network--specific metrics, the authors of [103] proposed a framework that uses common performance metrics, inspired by existing transportation systems. UAM network management concepts, namely slot-based, trajectory-based, and corridor-based approaches, have been provided.

4.3. Communication Reliability

Several technologies have been evaluated to determine the most effective communication link technologies for UAM. There is a real risk of collisions if links are dropped due to sudden high demand or if C2 links fail. Cellular networks need to be improved to make aerial vehicle communication more effective. Link disconnection problems can be mitigated by using redundant communication systems. Moreover, non-terrestrial and terrestrial network integration can provide better communication services for UAM applications.

4.4. Network Management

Researchers have made considerable efforts to develop 5G technology; however, communication services for UAM continue to be unsatisfactory. Three-dimensional movement in UAM, frequency handovers, and other networking factors make UAM communication more challenging [104,105]. A list of the networking and communication factors is presented in Table 7.

Table 7. Communication and networking challenges in UAM.

Technology	Advantages	Limitations
Cellular network	Currently, cellular networks, such as 5G, offer high data rates, low transmission delays, and support high mobility.	Due to the downward tilt of the antennas on existing mobile towers, they are unsuitable for UAM connectivity.
UAVs and HAPs	UAVs and HAPs offer large-area coverage and reliable communication and navigation for UAM.	The dense deployment of UAVs in urban areas poses a safety concern because of the possibility of collisions. HAPs links can be affected by rain attenuation.
LEO satellites	LEO satellites can provide near-continuous connectivity and navigation, and a long service time of up to 15 years.	Costly and require a large constellation, which leads to issues such as Doppler effects, fading, and rapid handovers.
NOMA	High-spectrum efficiency using the same resources (frequency) results in lower delays, because users are not required to wait for a slot before sending data.	A complex receiver and decoding process add additional delays, making it unsuitable for C2 links.
RSMA	Users can simultaneously send data by using multiple access techniques.	The complexity of the system increases with the number of users.
RIS	Increasing UAM connectivity through better coverage of BLOS communication in urban areas.	Channel state information is required to operate, but flying vehicles have a high level of mobility in a 3D environment. Further, RISs are subject to propagation and processing delays, which make them unsuitable for C2 communication.
Sidelink (D2D)	Allows distributed and decentralized multihop communication for flying vehicles. Moreover, sidelink provides a high data rate, throughput, and minimum delay.	Limited transmission range.
SDN	With SDN, UAM networks are highly scalable and easy to manage.	Virtualization of networks based on SDN faces security challenges.
mmWave	mmWave requires small antennae due to short-wave frequencies and allows higher bandwidth and data rates than RF.	Highly directional beamforming technology is required due to the strong attraction of high frequencies.
THz wave	High data rate and bandwidth.	Development is at an early stage and is limited by path loss and range.
Cloud and fog computing	UAM requires end-to-end path planning and weather information; in such cases flying vehicles utilize delay-tolerant clouds and fog computing for computation.	It is challenging to locate high-speed flying vehicles.
Edge computing	Edge computing is a distributed computing service that solves handover and offloading problems for flying vehicles.	A dense deployment of edge servers is required to provide this service.
Digital twin	Digital twins enable UAM simulations to optimize route planning, collision avoidance, and mission planning for UAM communication.	Currently, digital twins are not feasible for large-scale UAM networks because of their large data requirements.

4.5. Weather

Weather influence is a significant challenge in UAM operations. Aerial vehicles are subject to disruptions primarily caused by wind and in-path obstacles [106]. Weather conditions are more critical for UAM than for large aircraft. A variety of weather conditions can severely reduce the visibility of UAM, such as fog, haze, dust storms, rain, snow, and glare, making driving difficult. Sensor data are prone to disturbances, particularly during bad weather conditions. The UAM must be able to acquire and identify surrounding environmental information in real time and dynamically to make accurate judgments in different weather environments. Moreover, the flight status of other flying vehicles must be exchanged in real time. A UAM system's perception of the environment, whether good or bad, is comparable to that of a human, as it is related to the goal and objective of ensuring safe driving.

4.6. Noise

The success or failure of a UAM vehicle concept is also significantly influenced by the requirement for minimal noise emissions, which is a significant concern in terms of public acceptance. According to a survey on how the public perceives UAM, the majority expressed serious anxiety regarding the type and amount of noise generated by UAM. To the greatest extent feasible, UAM noise characteristics should be integrated into the current urban background soundscape to secure community acceptability. Blade tip speed is a crucial design factor that must be as low as is feasible. Potential solutions include spreading thrust generation across several rotors, reducing the weight, and increasing the rotor area. The latter runs counter to the requirement that aerial vehicles be as small as feasible to accommodate the constrained space and architecture in cities. Given the limited space available, this requirement suggests that aircraft size may be the limiting factor for the high data rate between UAM and its stations in urban areas [107].

4.7. Social Factors

Despite aiming to provide safe, sustainable, inexpensive, and accessible mobility, UAM must address concerns including societal equality, public acceptability of noise, and safety. UAM implementation may be made easier with demonstration projects, operating standards, limits on flying over residential areas at night, and inclement weather conditions. However, many challenges remain. Additionally, efforts should be made to examine the social and economic effects of UAM on local communities and to solve inequality issues. Testing for the integration of UAM in the same airspace as regular flights, safety and health effects, identification of data requirements, public perception, and best practices for multimodal integration and vertiport design are a few possible areas for further study. Moreover, further studies could concentrate on market entry strategies, commercial modeling, pertinent certification steps, the UAM environment, sensor and actuator developments, design challenges, technological requirements, and implications for the operational concept to drive UAM systems forward.

5. Conclusions

A revolutionary era of aeronautical communication began with the advent of UAM. In this article, we discussed the role of UAM in future mobility systems for the development of sustainable smart cities. Some key requirements must be satisfied for the successful deployment of UAM technologies. In particular, high-speed and high-accuracy data communication and mid-air vehicle-to-vehicle communication techniques are being developed. Simultaneously, each of these technologies requires high reliability and robustness. In densely populated areas, UAM systems require the overcoming of security and integration barriers, precise navigation and surveillance capabilities, vehicle-to-vehicle interactions, CDAA capabilities, and the management of communication failures. This paper discussed various aspects of UAM operational concepts and communication techniques. Moreover, our study analyzed in depth the key enabling technologies of UAM for communication and networking. Our study examined a number of challenges related to UAM development, including operational scenarios, ATM, routing, high-quality communication services, navigation, collision detection, path planning, weather effects, and noise control. At the end of the paper, we discussed future research directions for UAM deployment, and open research issues and challenges. Lastly, UAM can be successfully integrated into a smart living community when the public is involved, and most users consent to its daily use.

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